

MANPREET SINGH

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SOFTWARE DEVELOPER · FULL-STACK, FRONTEND, MOBILE

SUMMARY

Full-stack developer in Calgary with a Software Development diploma from SAIT (graduated August 2026), working in TypeScript, React and Next.js on the front and Node with PostgreSQL behind it. On a four-month capstone contract with a five-person team I delivered customer-facing features end to end on a production multi-tenant application and supported it after release. I have also led the Azure and Cosmos DB architecture for an edge-to-cloud system presented to industry guests, shipped a live AI web application, and built a cross-platform mobile app. Open to roles on-site in Calgary or remote across Canada, and eligible to work in Canada without sponsorship.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C#, SQL, Java, HTML, CSS, Bash

Frontend: React, Next.js 16 (App Router, server actions), React Native, Expo, Tailwind CSS, Framer Motion, responsive component-driven UI

Backend & APIs: Node.js, REST API design, Prisma ORM, .NET, .NET MAUI and Blazor Hybrid (Microsoft Learn certified), server-side authorization, role-based access control

Databases: PostgreSQL (Neon, Supabase), SQLite, Azure Cosmos DB, Cloud Firestore, schema design, migrations run as a deploy step, query optimization, indexing

Cloud & Infrastructure: Microsoft Azure (Azure Functions, Cosmos DB, AI Agent Service), Docker multi-stage builds, GitHub Actions CI/CD, Vercel, Firebase, Linux, AWS Cloud Practitioner (in progress)

AI Features Shipped: Model integration with schema-enforced structured output and fallback paths so a product degrades rather than fails, agent decision logic acting on live sensor data, MCP server integration

Testing & Quality: Vitest unit and integration tests against a live PostgreSQL instance in CI, test case design from requirements, defect tracking by severity and priority, regression testing

Practices: Agile delivery, Git branching, pull requests and peer code review, as-built and technical support documentation, UML class and sequence modelling, OOP, SDLC

EXPERIENCE

Full-Stack Developer, BAETT (BAETT-EMS)

May to Aug 2026

Calgary, AB · SAIT capstone contract · Part-time, 4 months · Team of 5 · Under NDA

- Delivered customer-facing features end to end on a production multi-tenant application, from requirements through unit testing and release, with peer code review on every pull request.
- Built authentication and role-based permissions, subscription billing covering payment gateway integration, plan modeling, checkout and usage tracking against customer limits, and secure document storage.
- Set up the CI pipeline running lint, type-check and automated tests against a live database before code merges, and wrote the internal documentation for the features I owned.

PROJECTS

RECEPIE: AI-Powered Pantry and Meal Planning Web Application

2026

Live: recepie-app-gilt.vercel.app · github.com/Manpreet445/Recepie

Tech: Next.js 16, TypeScript, React 19, Tailwind v4, Gemini API, Supabase (PostgreSQL), Framer Motion, Zod, Vitest

- Built a pantry-first meal planner: match ingredients on hand to recipes, generate multi-day AI meal plans, and consolidate everything into a single shopping list.
- Integrated the Gemini API with enforced JSON schemas plus a fallback data path, so the product stays usable when a model call fails; added a TDEE and macro calculator on Mifflin-St Jeor with unit tests over boundary cases.

AUTOMATED CONVEYOR SORTING SYSTEM: Cloud-Native Edge AI (Emergex)

Aug 2026

Tech: Azure, Azure Functions, Azure Cosmos DB, Azure AI Agent Service, Python, Raspberry Pi

- Cloud and Data Lead on a two-person build: designed the Azure backend ingesting classification events from an edge device rig and the Cosmos DB data model behind it.
- Implemented the AI agent decision logic translating sensor readings into sort actions on a live physical rig, and presented the architecture to industry guests at SAIT's INTP302 showcase.

STUDYSPOT: Real-Time Cross-Platform Mobile App

2026

github.com/Manpreet445/StudySpot · Tech: React Native, Expo SDK 55, Firebase (Firestore, Auth), Leaflet

- Built a crowd-sourced study spot finder from one Expo codebase serving iOS, Android and web, pushing live crowd-level updates to every connected client through Firestore listeners rather than polling, with maps resolved per platform.

EDUCATION

SAIT, Software Development Diploma · GPA 3.5 / 4.0

Jan 2025 to Aug 2026

Concurrent part-time work: Barista, Good Earth Coffeehouse · Crew Member, Firehouse Subs · 2025 to present